

Midhat Alvi

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EDUCATION

Rutgers University, School of Engineering

New Brunswick, NJ

Master of Engineering in Mechanical Engineering

GPA: 3.70 Jan 2026

Bachelor of Science in Mechanical Engineering (with Summa Cum Laude)

GPA: 3.66 May 2024

- Aerospace Engineering Concentration

WORK EXPERIENCE

Robotics Operations Engineer

Jan 2026 - Present

Human Friendly Robotics

- Authored the engineering plan, budget, and mechanical architecture for a \$3M, 36-month HUD grant proposal to demonstrate mobile robotics in factory-built housing.
- Designed custom body panels, brackets, and mounts in Fusion 360 and 3D printed them, assembling them with off-the-shelf structural components for an autonomous tile-laying construction robot to ensure fit, manufacturability, and serviceability.
- Integrated embedded electrical and control systems (RoboClaw motor controllers, Arduino, wireless communication, sensors, and power distribution) through wiring, soldering, and harness fabrication, to achieve reliable operation under real operating conditions.
- Designed and iterated on end-effector and material delivery subsystems, addressing real-world constraints such as alignment accuracy, flow consistency, and mechanical stability.
- Conducted system-level testing and validation, identifying and resolving issues related to drive performance, communication, and positioning under construction-site conditions.
- Supported pilot deployments and live customer demonstrations, troubleshooting in real time and translating contractor feedback into design improvements toward field readiness.

Thermal Engineer I

July 2025 - Dec 2025

Lummus Technology

- Supported thermal design of industrial fired heaters for \$15M-\$20M EPC projects, in accordance with customer specifications.
- Performed and verified thermal and mechanical checks including pressure drop, tube metal temperature (TMT), film temperature, and efficiency using in-house software and design tools.
- Applied API 530 / 560 and ASME Section I & VIII to validate tube wall thickness, material selection, and safe operating limits.
- Participated in formal design reviews, identifying design issues and contributing to full fired-heater designs impacting sizing, cost, procurement, and structural decisions.

PROJECTS

Bi-Propellant Liquid Rocket Engine

Senior Design/Capstone

- Performed thermal analysis for a 1,000 N thrust liquid rocket engine, informing combustion chamber material selection and high-temperature performance limits.

AIAA CLUBS

RU Airborne

Sep 2022 - May 2024

Propulsion Engineer - Build an RC plane for the AIAA Design Build Fly Competition

- Optimized aerodynamic properties through data analysis in Matlab, to meet competition requirements.
- Advanced team aerodynamics through research, wet layout, and rigorous tensile testing on carbon fiber wings, to reduce weight and boost structural strength.

SKILLS

- Technical Skills:** Fusion, Solidworks, Arduino Uno, Ansys, Microsoft Office, Simulink
- Coding Knowledge:** Matlab, Python, VBA
- Languages:** English (fluent), Urdu (fluent)